

KREMS, Austria

WMO: 110700

Lat: 48.4184N

Lon: 15.6216E

Elev: 666

StdP: 14.35

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.4	17.8	6.5	7.9	15.0	11.3	10.1	19.6	15.9	46.8	14.2	44.7	3.4	310	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	89.9	70.6	86.3	69.5	83.1	67.6	72.4	86.0	70.6	83.4	69.0	80.5	5.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	104.5	79.1	66.0	98.3	76.9	64.6	93.4	75.4	36.4	85.9	34.8	84.0	33.4	80.1	93.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.7	11.0	9.5	DB	8.3	96.0	6.5	4.8	3.6	99.4	-0.2	102.2	-3.8	104.9	-8.5	108.3	(4)
			WB	7.5	76.3	6.0	4.6	3.2	79.6	-0.3	82.3	-3.6	84.8	-8.0	88.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	32.3	35.4	42.7	52.6	59.6	67.1	70.0	68.9	59.9	50.3	42.2	34.3	(6)
(7)		DBStd	14.93	8.15	8.15	7.85	6.79	6.30	6.29	5.44	5.77	5.78	6.50	6.64	7.03	(7)
(8)		HDD50	2077	550	412	245	51	5	0	0	0	2	78	246	488	(8)
(9)		HDD65	5454	1014	830	693	374	187	49	16	26	172	456	684	953	(9)
(10)		CDD50	2573	1	2	17	129	303	514	621	587	300	88	11	0	(10)
(11)		CDD65	476	0	0	0	2	21	113	172	148	20	0	0	0	(11)
(12)		CDH74	4483	0	0	0	55	226	1050	1615	1340	189	8	0	0	(12)
(13)		CDH80	1464	0	0	0	8	36	330	572	483	35	0	0	0	(13)
(14)	Wind	WSAvg	4.1	4.4	4.3	4.7	4.6	4.4	4.1	4.1	3.6	3.5	3.4	3.8	3.9	(14)
(15)	Precipitation	PrecAvg	24.1	1.1	0.9	1.5	1.5	2.9	3.5	3.3	3.2	2.2	1.7	1.3	1.1	(15)
(16)		PrecMax	34.4	2.4	2.0	3.8	3.6	5.7	9.3	6.9	10.1	5.7	3.7	2.8	2.1	(16)
(17)		PrecMin	15.8	0.1	0.1	0.2	0.0	0.7	1.7	1.1	0.7	0.6	0.2	0.0	0.2	(17)
(18)		PrecStd	4.3	0.6	0.6	0.9	0.9	1.4	1.6	1.5	1.9	1.1	1.0	0.6	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	57.1	60.8	68.8	80.2	83.9	93.0	94.4	95.5	84.2	73.5	64.7	55.0	(19)	
(20)		MCWB	49.1	48.4	53.8	60.7	66.4	72.0	71.0	69.9	67.5	59.2	54.6	48.2	(20)	
(21)		DB	51.9	55.6	64.1	73.8	79.8	87.2	89.7	89.6	79.3	68.0	59.7	50.4	(21)	
(22)		MCWB	45.0	47.0	51.8	58.4	65.3	70.7	71.1	70.1	65.6	58.0	52.3	44.7	(22)	
(23)		DB	47.6	51.3	59.4	69.7	76.1	83.5	86.2	85.1	75.1	64.3	55.2	47.0	(23)	
(24)		MCWB	42.1	44.2	49.3	56.5	63.2	68.0	69.4	68.7	63.3	56.6	50.6	42.2	(24)	
(25)		DB	44.0	47.2	55.4	66.0	72.5	80.1	82.7	81.4	71.1	60.8	51.4	44.2	(25)	
(26)		MCWB	39.5	41.5	46.9	54.2	60.6	66.4	67.7	67.2	61.1	55.1	47.5	40.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	49.5	49.9	55.1	62.3	68.9	75.2	74.8	73.7	68.7	62.0	56.2	48.7	(27)	
(28)		MCDWB	56.2	58.6	66.8	78.2	80.8	85.8	88.7	88.2	81.2	69.4	62.2	54.6	(28)	
(29)		WB	45.5	47.1	52.2	59.2	66.5	72.0	72.2	71.4	66.5	59.9	53.4	45.2	(29)	
(30)		MCDWB	51.5	55.0	62.3	72.5	77.8	84.2	86.1	85.7	77.6	66.1	58.0	49.6	(30)	
(31)		WB	42.5	44.6	50.2	57.1	64.3	69.6	70.6	69.7	64.5	57.6	50.6	42.6	(31)	
(32)		MCDWB	47.1	51.4	58.3	67.7	74.1	80.6	83.4	82.8	73.7	62.7	54.9	46.3	(32)	
(33)		WB	39.9	41.8	47.7	55.2	61.9	67.5	68.9	67.9	62.2	55.7	48.2	40.6	(33)	
(34)		MCDWB	43.7	46.7	54.8	65.0	70.8	77.7	80.2	79.5	69.4	60.4	51.0	43.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.3	12.8	16.6	19.8	18.6	19.0	19.6	20.3	18.0	15.6	10.9	10.0	(35)	
(36)		MCDBR	16.0	20.1	25.9	27.0	25.4	25.4	26.3	27.4	25.2	21.7	17.7	15.8	(36)	
(37)		MCWBR	10.9	12.8	15.1	13.7	12.7	11.3	10.6	11.5	12.7	13.1	11.5	11.2	(37)	
(38)		MCDBR	15.6	19.5	24.4	25.0	24.0	23.1	23.5	25.0	23.9	19.4	16.2	14.8	(38)	
(39)	MCWBR	10.9	12.8	14.7	13.2	12.6	11.0	10.4	11.0	12.3	12.3	10.9	10.9	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.310	0.335	0.375	0.418	0.420	0.425	0.432	0.429	0.400	0.368	0.340	0.309	(40)	
(41)		taud	2.387	2.346	2.280	2.200	2.238	2.261	2.243	2.268	2.316	2.380	2.413	2.427	(41)	
(42)		Ebn at Noon	239	258	264	263	267	266	262	257	253	242	225	224	(42)	
(43)		Edh at Noon	22	28	35	42	42	42	42	39	34	27	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	318	596	936	1370	1588	1734	1677	1496	1068	666	341	252	(44)	
(45)		RadStd	48	78	120	163	183	150	162	151	137	76	36	28	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (92 neighbors)	+0.95	N/A	N/A	N/A	+0.72	+0.61	-212	-270	+131	+59	(47)

Nomenclature: See separate page